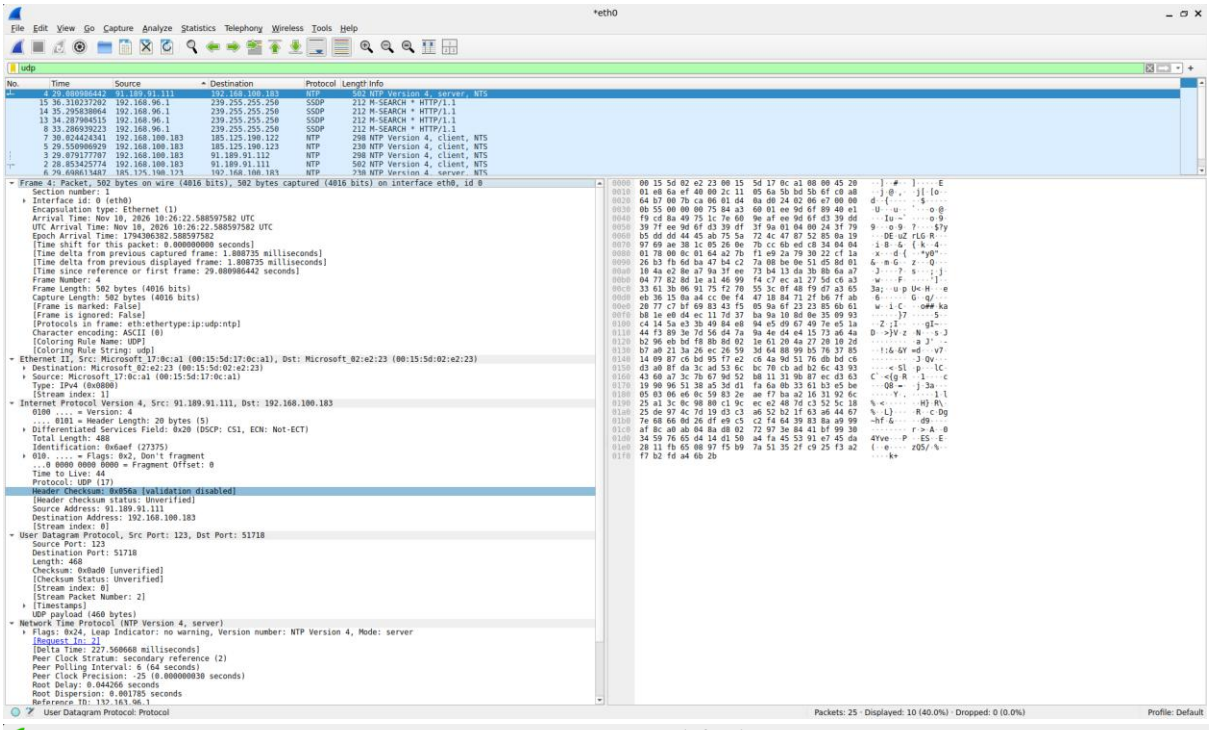
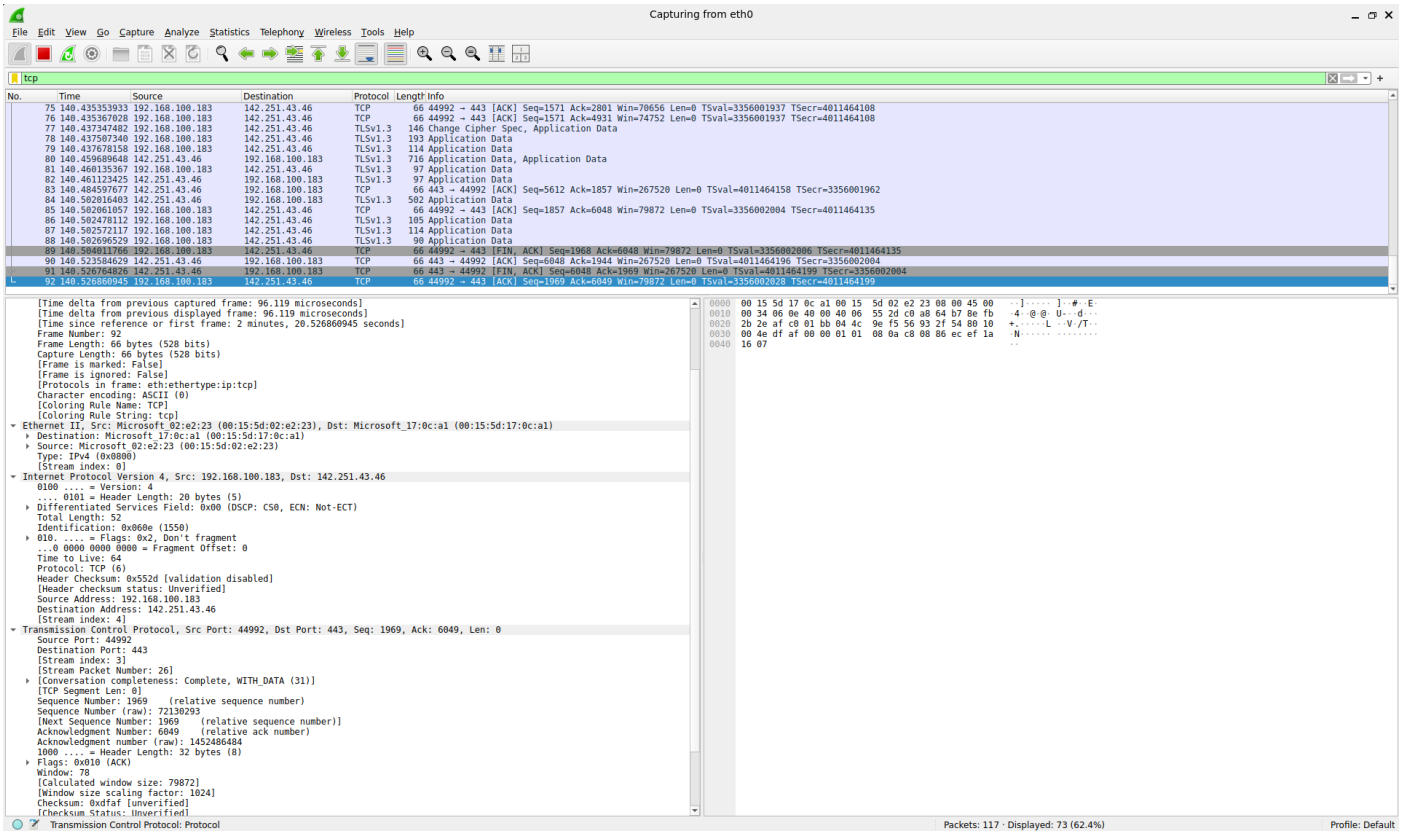

*eth0
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
udp
No. Time Source Destination Protocol Length Info
4 29.080986442 91.189.91.111 192.168.100.183 NTP 502 NTP Version 4, server
15 36.310237202 192.168.96.1 239.255.255.250 SSDP 212 M-SEARCH * HTTP/1.1
14 35.295838064 192.168.96.1 239.255.255.250 SSDP 212 M-SEARCH * HTTP/1.1
13 34.287904515 192.168.96.1 239.255.255.250 SSDP 212 M-SEARCH * HTTP/1.1
8 33.286939223 192.168.96.1 239.255.255.250 SSDP 212 M-SEARCH * HTTP/1.1
7 30.024424341 192.168.100.183 185.125.190.122 NTP 298 NTP Version 4, client, NTS
5 29.550906929 192.168.100.183 185.125.190.123 NTP 230 NTP Version 4, client, NTS
3 29.079177707 192.168.100.183 91.189.91.112 NTP 298 NTP Version 4, client, NTS
2 28.853425774 192.168.100.183 91.189.91.111 NTP 502 NTP Version 4, client, NTS
6 29.460615487 185.125.190.123 192.168.100.183 NTP 230 NTP Version 4, server, NTS
Frame 4: Packet, 502 bytes on wire (4016 bits), 502 bytes captured (4016 bits) on interface eth0, id 0
Section number: 1
Interface id: 0 (eth0)
Encapsulation type: Ethernet (1)
Arrival Time: Nov 10, 2026 10:26:22.588597582 UTC
UTC Arrival Time: Nov 10, 2026 10:26:22.588597582 UTC
Epoch Arrival Time: 1794306382.588597582
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 1.808735 milliseconds]
[Time delta from previous displayed frame: 1.808735 milliseconds]
[Time since reference or first frame: 29.080986442 seconds]
Frame Number: 4
Frame Length: 502 bytes (4016 bits)
Capture Length: 502 bytes (4016 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:udp:ntp]
Character encoding: ASCII (0)
[Coloring Rule Name: UDP]
[Coloring Rule String: udp]
Ethernet II, Src: Microsoft_17:0c:a1 (00:15:5d:17:0c:a1), Dst: Microsoft_02:e2:23 (00:15:5d:02:e2:23)
Destination: Microsoft_02:e2:23 (00:15:5d:02:e2:23)
Source: Microsoft_17:0c:a1 (00:15:5d:17:0c:a1)
Type: IPv4 (0x0800)
[Stream index: 1]
Internet Protocol Version 4, Src: 91.189.91.111, Dst: 192.168.100.183
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x20 (DSCP: CS1, ECN: Not-ECT)
Total Length: 488
Identification: 0x6aef (27375)
010. .... = Flags: 0x2, Don't fragment
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 44
Protocol: UDP (17)
Header Checksum: 0x856a [validation disabled]
[Header checksum status: Unverified]
Source Address: 91.189.91.111
Destination Address: 192.168.100.183
[Stream index: 0]
User Datagram Protocol, Src Port: 123, Dst Port: 51718
Source Port: 123
Destination Port: 51718
Length: 468
Checksum: 0x8ad0 [unverified]
[Checksum Status: Unverified]
[Stream index: 0]
[Stream Packet Number: 2]
[Timestamps]
UDP payload (460 bytes)
Network Time Protocol (NTP Version 4, server)
Flags: 0x24, Leap Indicator: no warning, Version number: NTP Version 4, Mode: server
[Request In: 2]
[Delta Time: 227.560668 milliseconds]
Peer Clock Stratum: secondary reference (2)
Peer Polling Interval: 6 (64 seconds)
Peer Clock Precision: -25 (0.000000030 seconds)
Root Delay: 0.044266 seconds
Root Dispersion: 0.001785 seconds
Reference ID: 132.163.96.1
User Datagram Protocol: Protocol
Packets: 25 · Displayed: 10 (40.0%) · Dropped: 0 (0.0%)
Profile: Default


Capturing from eth0
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp
No. Time Source Destination Protocol Length Info
75 140.435353933 192.168.100.183 142.251.43.46 TCP 66 44992 → 443 [ACK] Seq=1571 Ack=2801 Win=70656 Len=0 TSval=3356001937 TSecr=4011464108
76 140.435367028 192.168.100.183 142.251.43.46 TCP 66 44992 → 443 [ACK] Seq=1571 Ack=4931 Win=74752 Len=0 TSval=3356001937 TSecr=4011464108
77 140.437347482 192.168.100.183 142.251.43.46 TLSv1.3 146 Change Cipher Spec, Application Data
78 140.437507340 192.168.100.183 142.251.43.46 TLSv1.3 193 Application Data
79 140.437678158 192.168.100.183 142.251.43.46 TLSv1.3 114 Application Data
80 140.459689648 142.251.43.46 192.168.100.183 TLSv1.3 716 Application Data, Application Data
81 140.460135367 192.168.100.183 142.251.43.46 TLSv1.3 97 Application Data
82 140.461123425 142.251.43.46 192.168.100.183 TLSv1.3 97 Application Data
83 140.484597677 142.251.43.46 192.168.100.183 TCP 66 443 → 44992 [ACK] Seq=5612 Ack=1857 Win=267520 Len=0 TSval=4011464158 TSecr=3356001962
84 140.502016403 142.251.43.46 192.168.100.183 TLSv1.3 502 Application Data
85 140.502061057 192.168.100.183 142.251.43.46 TCP 66 44992 → 443 [ACK] Seq=1857 Ack=6048 Win=79872 Len=0 TSval=3356002004 TSecr=4011464135
86 140.502478112 192.168.100.183 142.251.43.46 TLSv1.3 105 Application Data
87 140.502572117 192.168.100.183 142.251.43.46 TLSv1.3 114 Application Data
88 140.502696529 192.168.100.183 142.251.43.46 TLSv1.3 90 Application Data
89 140.504811766 192.168.100.183 142.251.43.46 TCP 66 44992 → 443 [FIN, ACK] Seq=1968 Ack=6048 Win=79872 Len=0 TSval=3356002006 TSecr=4011464135
90 140.523584629 142.251.43.46 192.168.100.183 TCP 66 443 → 44992 [ACK] Seq=6048 Ack=1944 Win=267520 Len=0 TSval=4011464196 TSecr=3356002004
91 140.526764826 142.251.43.46 192.168.100.183 TCP 66 443 → 44992 [FIN, ACK] Seq=6048 Ack=1969 Win=267520 Len=0 TSval=4011464199 TSecr=3356002004
92 140.526860945 192.168.100.183 142.251.43.46 TCP 66 44992 → 443 [ACK] Seq=1969 Ack=6049 Win=79872 Len=0 TSval=3356002028 TSecr=4011464199
[Time delta from previous captured frame: 96.119 microseconds]
[Time delta from previous displayed frame: 96.119 microseconds]
[Time since reference or first frame: 2 minutes, 20.526860945 seconds]
Frame Number: 92
Frame Length: 66 bytes (528 bits)
Capture Length: 66 bytes (528 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:tcp]
Character encoding: ASCII (0)
[Coloring Rule Name: TCP]
[Coloring Rule String: tcp]
Ethernet II, Src: Microsoft_02:e2:23 (00:15:5d:02:e2:23), Dst: Microsoft_17:0c:a1 (00:15:5d:17:0c:a1)
Destination: Microsoft_17:0c:a1 (00:15:5d:17:0c:a1)
Source: Microsoft_02:e2:23 (00:15:5d:02:e2:23)
Type: IPv4 (0x0800)
[Stream index: 0]
Internet Protocol Version 4, Src: 192.168.100.183, Dst: 142.251.43.46
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 52
Identification: 0x060e (1550)
010. .... = Flags: 0x2, Don't fragment
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 64
Protocol: TCP (6)
Header Checksum: 0x552d [validation disabled]
[Header checksum status: Unverified]
Source Address: 192.168.100.183
Destination Address: 142.251.43.46
[Stream index: 4]
Transmission Control Protocol, Src Port: 44992, Dst Port: 443, Seq: 1969, Ack: 6049, Len: 0
Source Port: 44992
Destination Port: 443
[Stream index: 3]
[Stream Packet Number: 26]
[Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 0]
Sequence Number: 1969 (relative sequence number)
Sequence Number (raw): 72130293
[Next Sequence Number: 1969 (relative sequence number)]
Acknowledgment Number: 6049 (relative ack number)
Acknowledgment number (raw): 1452486484
1000 .... = Header Length: 32 bytes (8)
Flags: 0x010 (ACK)
Window: 78
[Calculated window size: 79872]
[Window size scaling factor: 1024]
Checksum: 0xdfaf [unverified]
[Checksum Status: Unverified]
Transmission Control Protocol: Protocol
Packets: 117 · Displayed: 73 (62.4%)
Profile: Default

Completed Observation Table

Parameter	TCP Packet (Frame 92)	UDP Packet (Frame 24)
Frame Length	66 bytes	238 bytes
Ethernet Header Length	14 bytes	14 bytes
IP Header Length	20 bytes	20 bytes
IP Total Length	52 bytes	216 bytes
Transport Header Length	32 bytes *(20B base + 12B TCP options like TSval)*	8 bytes *(fixed standard UDP header)*
Payload Length	0 bytes *(pure ACK control frame, Len=0)*	188 bytes *(NTP payload)*

Header Overhead & Efficiency Calculations

1. UDP Packet (Frame 24 with data)

- **Total Headers Overhead:**

$$\text{Ethernet (14B)} + \text{IPv4 (20B)} + \text{UDP (8B)} = 42 \text{ bytes}$$

- **Application Data (Payload):**

$$188 \text{ bytes}$$

- **Total Frame Size:**

$$188 + 42 = 230 \text{ bytes}$$

(excluding 4-byte FCS not reported by OS)

- **Payload Efficiency:**

$$\frac{\text{Payload Length}}{\text{Total Frame Length}} \times 100 = \frac{188}{238} \times 100 \approx 78.99\%$$

2. Theoretical Full-Sized TCP Packet (from manual's benchmark)

- **Total Overhead:**

$$14 \text{ (Ethernet)} + 20 \text{ (IP)} + 20 \text{ (TCP)} + 4 \text{ (FCS)} = 58 \text{ bytes}$$

- **Efficiency with 1500-byte Payload:**

$$\frac{1500}{1558} \times 100 \approx 96.28\%$$

Answers to the Lab Questions

1. What is encapsulation?

Encapsulation is the process where each layer of the OSI or TCP/IP protocol stack wraps lower-layer payload data with its own protocol-specific header (and sometimes trailer) before handing it down to the next physical transmission stage.

2. What is the purpose of the Ethernet header?

It provides Data Link Layer (Layer 2) addressing via Source and Destination MAC addresses and identifies the encapsulated Layer 3 protocol (via the EtherType field, e.g., 0x0800 for IPv4) to direct the frame across the local physical network.

3. What information is stored in the IP header?

The IPv4 header contains Source and Destination IP addresses, IP Version, Header Length (IHL), Total Packet Length, Time to Live (TTL), Protocol identifier (e.g., TCP=6, UDP=17), Fragmentation flags/offsets, and a Header Checksum.

4. What is the difference between TCP and UDP headers?

- **TCP Header:** A complex, connection-oriented header with a base size of 20 bytes (expandable via options) containing sequence numbers, acknowledgment numbers, control flags (SYN, ACK, FIN, RST), window size, and checksum for reliable transmission.
- **UDP Header:** A minimalist, connectionless header fixed at only 8 bytes containing only Source Port, Destination Port, Length, and Checksum for low-latency transmission.

5. Why does each protocol add a header?

Each protocol layer provides distinct network functions (Layer 2 handles local physical addressing, Layer 3 handles global logical routing, Layer 4 handles process-to-process multiplexing and reliability). A separate header provides the control data necessary for that specific layer to deliver and decode the transmission.

6. What is packet overhead?

Packet overhead is the total bandwidth and byte space consumed by protocol headers (Ethernet, IP, TCP/UDP) and frame check sequences rather than the actual user or application payload.

7. How does header overhead affect network efficiency?

Header overhead reduces overall network efficiency by consuming link bandwidth with non-payload metadata. For large payloads (such as file downloads), efficiency is high ($\approx 96\%$), whereas for very small messages or keep-alive packets, headers can make up the vast majority of the transmitted frame, significantly reducing protocol efficiency.